Supplementary Safe-RL Results

We evaluate a practical neural-network instantiation of REP-PD-TEG, denoted `REP-PD-NN`. It replaces the theoretical per-round full-batch MLE representation step with a mini-batch SPEDER-style stochastic representation objective [1], uses the learned state-action representation in neural reward and cost critics, and trains the policy with a Lagrangian-style constrained policy update. We compare it with a `raw` baseline that uses the same Lagrangian training skeleton but removes the representation learner.

Safety Gym experiments. The table reports results on three Safety Gym continuous-control environments. All rows use 1M interaction steps, 3 random seeds, the safe RL training hyperparameters specified in the reproduction script, and the final deterministic evaluation checkpoint.

Environment	Raw Ret.	Raw Cost	REP-PD-NN Ret.	REP-PD-NN Cost
PointGoal1	-1.38 ± 0.58	27.85 ± 18.99	$\mathbf{-0.14 \pm 0.03}$	$\mathbf{16.45 \pm 12.69}$
PointPush1	-1.40 ± 0.60	32.17 ± 14.66	$\mathbf{-0.17 \pm 0.12}$	$\mathbf{12.13 \pm 6.95}$
PointPush2	-1.40 ± 0.71	13.00 ± 8.17	$\mathbf{-0.03 \pm 0.01}$	$\mathbf{10.90 \pm 3.85}$

Across these three tasks, `REP-PD-NN` improves final return and reduces final cost, suggesting that learned representations can be beneficial for practical safe RL in continuous-control settings.

Reference. [1] T. Ren et al., Spectral Decomposition Representation for Reinforcement Learning.